

AP PRECALCULUS · THE UNIVERSAL STUDY GUIDE

3 Units · 4 FRQ Types · Function-Centered Thinking · The Score-Defining Moves

Tag legend: **[FLOOR—4]** = must-have **[STRETCH—5]** = what gets you the 5

PAGE 1 | MATH OF YOUR TARGET · THE 4 FRQ TYPES · CALCULATOR STRATEGY

THE MATH OF YOUR TARGET — what you actually need to score (~75%)

Section	Weight	Questions	Get this many	Time
MCQ Part A (no calculator)	43.75%	28 questions	21 / 28 (~75%)	80 min · ~3 min/Q
MCQ Part B (calculator)	18.75%	12 questions	9 / 12 (~75%)	40 min · ~3.3 min/Q
FRQ Part A (calculator)	18.75%	2 questions	~80% of the points	30 min · 15 min each
FRQ Part B (no calculator)	18.75%	2 questions	~80% of the points	30 min · 15 min each

Note: Only Units 1, 2, and 3 are tested. **Unit 4 (parameters/vectors/matrices) is NOT on the AP exam** — don't waste study time there.

THE 4 FRQ TYPES — they are FIXED every year. Prep for each specifically.

FRQ	Type (Calculator?)	What it asks + the score-earning move
FRQ 1	Function Concepts (calculator)	Analyze a function (graph/table/equation): find zeros, extrema, intervals of inc/dec, concavity, average rates of change. Score-earning move: use the exact terminology the prompt asks for and show work even when calculator gives the answer .
FRQ 2	Modeling Non-Periodic (calculator)	Real-world scenario fitting linear/quadratic/exponential/log model. Score-earning move: identify the function family from the data (constant differences = linear; constant ratios = exponential; constant 2nd differences = quadratic), then use technology to fit, then INTERPRET in context.
FRQ 3	Modeling Periodic (no calculator)	Sinusoidal model from real-world data. Score-earning move: extract amplitude, midline, period, phase shift directly from given max/min/period data. Choose $\sin$ or $\cos$ based on starting position.
FRQ 4	Symbolic Manipulations (no calculator)	Pure algebra: solve, factor, simplify, prove identities, manipulate expressions. Score-earning move: show every algebraic step — the rubric awards setup points even before the final answer.

CALCULATOR STRATEGY — when to use it, when to verify by hand [5]

Calculator usage rules on the exam: allowed on MCQ Part B + FRQ Part A only. Use a graphing calculator with these 4 capabilities cold:

1. **Graph a function** — visualize end behavior, intercepts, asymptotes.
2. **Find zeros / x-intercepts** (Calc menu → zero feature).
3. **Find max/min** (Calc → maximum/minimum) — check local vs absolute.
4. **Find intersections** (Calc → intersect) — for solving equations graphically.
5. **Evaluate a function at a point** (table or $Y_1(x)$ syntax).

[5] CRITICAL — Mode check before Section 1: Set calculator to **RADIAN** mode for trig. AP Precalc uses radians by default. **Most missed-by-1-point errors** come from degree mode.

THE 4-vs-5 DIFFERENTIATOR (the central insight)

A 4-scorer gets the answer. A 5-scorer SHOWS WORK + JUSTIFIES + USES PROPER NOTATION. The College Board's FRQ rubric is unforgiving — a correct answer with no work shown often earns only 1 of 3 points. **Always: setup → algebra → answer → context.**

4-scorer: " $x = 4$."

5-scorer: "Setting the function equal to zero: $x^2 - 5x + 4 = 0 \rightarrow$ factoring: $(x-4)(x-1) = 0 \rightarrow$ solutions: $x = 1$ and $x = 4$. The function has zeros at $x = 1$ and $x = 4$. **In context, these represent the times at which the population returns to its initial value.**"

PAGE 2 | MCQ MASTERY + FUNCTION CONCEPTS · the universal exam vocabulary

MCQ STRATEGY — eliminate, don't always solve [4]

~3 min per question is plenty IF you're efficient. Most MCQs reward recognition of FORM, not raw computation.

- **Plug in test values** when stuck on "which expression equals..." Pick $x = 0$, $x = 1$, or $x = 2$ and evaluate both the question and each option.
- **Eliminate by domain.** If the answer must be positive but option C is negative, eliminate.
- **Eliminate by end behavior.** Polynomial degree + leading coefficient sign tells you which way the function goes — wrong end behavior = wrong answer.
- **Eliminate by transformations.** $f(x - 3)$ shifts RIGHT 3, not left. Wrong direction = eliminate.
- **For inverse questions,** swap x and y mentally. Inverse of $(2, 5)$ on the original is $(5, 2)$ on the inverse.
- **Save calculator MCQs for last** — front-load no-calc questions where you have momentum.

THE FUNCTION CONCEPTS FRAMEWORK — every FRQ uses this language [5]

This is the vocabulary of the exam. Use these EXACT terms in your FRQ answers — graders look for them.

Concept	What it means + how to find it
Domain	Set of valid x -inputs. Excluded: division by 0, log of non-positive, even root of negative.
Range	Set of possible y -outputs. Read off graph's vertical extent.
Zeros / x-intercepts	Values of x where $f(x) = 0$. Solve algebraically OR use calculator zero feature.
y-intercept	Value of $f(0)$. Plug in 0.
Increasing / Decreasing	Increasing on (a,b) means: for any $x_1 < x_2$ in (a,b) , $f(x_1) < f(x_2)$. Look at graph shape.
Concave up / down	Concave up: rate of change is increasing — graph curls upward (cup shape). Concave down: rate decreasing — graph curls downward (frown shape).
End behavior	What $f(x)$ does as $x \rightarrow \infty$ or $x \rightarrow -\infty$. For polynomials: degree + leading coefficient sign decides.
Local max / min	Highest/lowest point in a neighborhood (not necessarily overall).
Absolute (global) max / min	Highest/lowest point in the ENTIRE domain.
Asymptote (vertical / horizontal / slant)	Line the graph approaches but never reaches. Vertical at zeros of denominator (rational); horizontal from degree comparison.
Average rate of change over $[a, b]$	$\frac{f(b) - f(a)}{b - a}$ — the slope of the secant line. THE precalc analog of derivative.
Symmetry	Even: $f(-x) = f(x)$ (y -axis symmetry). Odd: $f(-x) = -f(x)$ (origin symmetry).

AVERAGE RATE OF CHANGE — your most common FRQ tool [5]

$$\text{AVG rate of change of } f \text{ over } [a, b] = \frac{f(b) - f(a)}{b - a}$$

- Same formula as 'slope' but for any function over an interval.
- If avg ROC over $[a, b]$ is positive $\Rightarrow f$ is INCREASING on average
- If avg ROC > 0 AND increasing as interval shifts right $\Rightarrow f$ is CONCAVE UP

Always show the formula setup, then plug in. "Average rate of change of f over $[2, 5] = \frac{f(5) - f(2)}{5 - 2} = \frac{(12 - 4)}{3} = 8/3$."

FUNCTION TRANSFORMATIONS — the cheat strip [4]

Starting with $y = f(x)$:

- $y = f(x) + k \rightarrow$ shift UP by k (vertical shift)
- $y = f(x - h) \rightarrow$ shift RIGHT by h (horizontal shift) ← note: opposite of sign
- $y = a \cdot f(x) \rightarrow$ vertical stretch by $|a|$; if $a < 0$, reflect over x -axis
- $y = f(b \cdot x) \rightarrow$ horizontal compression by $1/|b|$; if $b < 0$, reflect over y -axis
- $y = -f(x) \rightarrow$ reflect over the x -axis (flip vertically)
- $y = f(-x) \rightarrow$ reflect over the y -axis (flip horizontally)

UNIT 1 | POLYNOMIAL & RATIONAL FUNCTIONS

Polynomial Behavior — degree + leading coefficient = end behavior

General polynomial: $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$

END BEHAVIOR depends on degree (n) and lead coefficient (a_n):

Even n , + lead	$\uparrow$ as $x \rightarrow -\infty$, $\uparrow$ as $x \rightarrow +\infty$	(both ends UP)
Even n , - lead	$\downarrow$ as $x \rightarrow -\infty$, $\downarrow$ as $x \rightarrow +\infty$	(both ends DOWN)
Odd n , + lead	$\downarrow$ as $x \rightarrow -\infty$, $\uparrow$ as $x \rightarrow +\infty$	(down-up — left lower)
Odd n , - lead	$\uparrow$ as $x \rightarrow -\infty$, $\downarrow$ as $x \rightarrow +\infty$	(up-down)

Zeros and Multiplicity — what the graph does AT each zero

- **Multiplicity 1 (odd):** graph CROSSES the x -axis straight through. $(x - 2)^1 \rightarrow$ zero at $x=2$, crosses.
- **Multiplicity 2 (even):** graph TOUCHES and bounces off. $(x - 2)^2 \rightarrow$ zero at $x=2$, bounces.
- **Multiplicity 3 (odd):** graph crosses but FLATTENS through. $(x - 2)^3 \rightarrow$ zero at $x=2$, flattens.
- **Sum of multiplicities = degree of the polynomial.** $(x-1)^2(x+3)(x-5)^3$ has multiplicities $2+1+3 = 6 \rightarrow$ degree 6.

Rational Functions — asymptote rules [5]

For $f(x) = N(x) / D(x)$ where N and D are polynomials:

VERTICAL asymptote: at $x = c$ where $D(c) = 0$ AND $N(c) \neq 0$

HOLE (removable): at $x = c$ where BOTH $N(c) = 0$ AND $D(c) = 0$ (factor cancels)

HORIZONTAL asymptote: depends on degrees of N and D

- $\deg(N) < \deg(D) \rightarrow y = 0$
- $\deg(N) = \deg(D) \rightarrow y = (\text{lead coeff of } N) / (\text{lead coeff of } D)$
- $\deg(N) > \deg(D) \rightarrow$ no horizontal asymptote

SLANT (oblique) asymptote: when $\deg(N) = \deg(D) + 1$

$\rightarrow$ do polynomial long division; quotient = slant asymptote

Inverse Functions — when they exist + how to find [4]

- **Inverse exists if function is one-to-one** (passes the **horizontal line test**).
- **To find inverse algebraically:** Replace $f(x)$ with y . **Swap x and y .** Solve for new y . Replace y with $f^{-1}(x)$.
- **Graph of f^{-1}** is the reflection of f across the line $y = x$.
- **Composition check:** $f(f^{-1}(x)) = x$ AND $f^{-1}(f(x)) = x$.

Polynomial Operations & Equivalent Forms [5]

- **Factor form:** $P(x) = a(x - r_1)(x - r_2)\dots$ — reveals zeros.
- **Standard form:** $P(x) = a_n x^n + \dots + a_0$ — reveals end behavior.
- **Vertex form for quadratic:** $P(x) = a(x - h)^2 + k$ — reveals max/min at (h, k) .
- **Polynomial long division** — needed for slant asymptotes + factoring high-degree polynomials.
- **Synthetic division** — shortcut when dividing by $(x - c)$. Faster for testing zeros.

Even & Odd Functions [4]

- **Even:** $f(-x) = f(x)$ — symmetric about y -axis. Examples: x^2 , $\cos x$, $|x|$.
- **Odd:** $f(-x) = -f(x)$ — symmetric about origin. Examples: x^3 , $\sin x$, $\tan x$.
- **Most polynomials are neither.** Test by plugging in $-x$ and comparing.

UNIT 2 | EXPONENTIAL & LOGARITHMIC FUNCTIONS

Exponential Functions — the foundation [4]

General form: $f(x) = a \cdot b^x$ (a = initial value, b = base)

Growth: $b > 1$ (e.g., $b = 2, 1.05, e \approx 2.718$)

Decay: $0 < b < 1$ (e.g., $b = 0.5, 0.95$)

Horizontal asymptote: $y = 0$ (always, for unshifted exponentials)

y-intercept: $f(0) = a$

Continuous compounding: $A = P \cdot e^{(rt)}$ (r = rate, t = time)

Periodic compounding: $A = P \cdot (1 + r/n)^{(nt)}$ (n = compoundings per year)

Logarithm Properties — drill the 5 essentials [5]

PRODUCT: $\log(M \cdot N) = \log M + \log N$

QUOTIENT: $\log(M / N) = \log M - \log N$

POWER: $\log(M^k) = k \cdot \log M$

ZERO: $\log_b(1) = 0$

IDENTITY: $\log_b(b) = 1$

CHANGE OF BASE: $\log_b(x) = \log(x) / \log(b) = \ln(x) / \ln(b)$

INVERSE relationship: $b^x = a \Leftrightarrow \log_b(a) = x$

Common shorthand: $\log x = \log_{10} x \quad | \quad \ln x = \log_e x$

Solving Exponential Equations [5]

- **Same base?** Set exponents equal. $2^{(x+1)} = 2^5 \Rightarrow x + 1 = 5 \Rightarrow x = 4$.
- **Different bases?** Take log (or ln) of both sides. $3^x = 17 \Rightarrow x \cdot \log 3 = \log 17 \Rightarrow x = \log 17 / \log 3$.
- **Quadratic in disguise:** $e^{(2x)} - 5e^x + 6 = 0 \Rightarrow$ let $u = e^x \Rightarrow u^2 - 5u + 6 = 0 \Rightarrow$ factor $\rightarrow$ solve $\rightarrow$ back-substitute.

Solving Logarithmic Equations [5]

- **Combine logs first** using product/quotient/power properties to a single log.
- **Convert to exponential form** to solve. $\log_b(M) = N \Rightarrow M = b^N$.
- **ALWAYS check for extraneous solutions.** Logs are only defined for **positive** arguments — solutions making any log argument ≤ 0 must be rejected.

Identifying the Right Model from Data [5]

This is the score-defining move on FRQ 2 (Modeling Non-Periodic). Examine the data:

If equally-spaced x-values yield...	Then the model is...	Why
Constant first differences (e.g., y values increase by 3, 3, 3, ...)	Linear: $y = mx + b$	Constant rate of change = constant slope
Constant second differences	Quadratic: $y = ax^2 + bx + c$	Slope of slope is constant
Constant ratios (y values multiply by 1.5, 1.5, 1.5, ...)	Exponential: $y = a \cdot b^x$	Constant percentage change
Linear when plotted on semi-log axis	Exponential: $y = a \cdot b^x$	Logarithm of exponential is linear

Exponential vs Linear Modeling — the conceptual distinction [5]

- **Linear:** "increases by 5 each year" (constant absolute change)
- **Exponential:** "increases by 5% each year" (constant percentage change)
- **Half-life / doubling time:** characteristic of exponential. Find from b : doubling time $= \log 2 / \log b$.

UNIT 3 | TRIGONOMETRIC FOUNDATIONS · UNIT CIRCLE · IDENTITIES

THE UNIT CIRCLE — memorize cold (no calc on Part A & FRQ 3, 4) [5]

Each angle gives a point $(\cos \theta, \sin \theta)$ on the unit circle:

ANGLE	DEG	RAD	$(\cos \theta, \sin \theta)$
0°	0	0	(1, 0)
30°	30	$\pi/6$	$(\sqrt{3}/2, 1/2)$
45°	45	$\pi/4$	$(\sqrt{2}/2, \sqrt{2}/2)$
60°	60	$\pi/3$	$(1/2, \sqrt{3}/2)$
90°	90	$\pi/2$	(0, 1)
120°	120	$2\pi/3$	$(-1/2, \sqrt{3}/2)$
135°	135	$3\pi/4$	$(-\sqrt{2}/2, \sqrt{2}/2)$
150°	150	$5\pi/6$	$(-\sqrt{3}/2, 1/2)$
180°	180	π	(-1, 0)
210°	210	$7\pi/6$	$(-\sqrt{3}/2, -1/2)$
225°	225	$5\pi/4$	$(-\sqrt{2}/2, -\sqrt{2}/2)$
240°	240	$4\pi/3$	$(-1/2, -\sqrt{3}/2)$
270°	270	$3\pi/2$	(0, -1)
300°	300	$5\pi/3$	$(1/2, -\sqrt{3}/2)$
315°	315	$7\pi/4$	$(\sqrt{2}/2, -\sqrt{2}/2)$
330°	330	$11\pi/6$	$(\sqrt{3}/2, -1/2)$

Memory hack: Quadrant signs follow "All Students Take Calculus" — Q1 All +, Q2 Sin +, Q3 Tan +, Q4 Cos +. ($\sin = y$, $\cos = x$)

The Six Trig Functions — definitions on the unit circle [5]

$$\begin{aligned}\sin \theta &= y & \csc \theta &= 1/\sin \theta = 1/y \\ \cos \theta &= x & \sec \theta &= 1/\cos \theta = 1/x \\ \tan \theta &= y/x & \cot \theta &= 1/\tan \theta = x/y \quad (= \cos \theta / \sin \theta)\end{aligned}$$

Domain notes: $\tan$ & $\sec$ undefined where $x = 0$ (i.e., $\theta = \pi/2, 3\pi/2$)
 $\cot$ & $\csc$ undefined where $y = 0$ (i.e., $\theta = 0, \pi$)

Periods: $\sin, \cos, \sec, \csc \rightarrow 2\pi$ $\tan, \cot \rightarrow \pi$

Trig Identities — the must-know set [5]

PYTHAGOREAN identities (one + two derivatives):

$$\begin{aligned}\sin^2 \theta + \cos^2 \theta &= 1 \\ 1 + \tan^2 \theta &= \sec^2 \theta && \text{(divide first by } \cos^2 \text{)} \\ \cot^2 \theta + 1 &= \csc^2 \theta && \text{(divide first by } \sin^2 \text{)}\end{aligned}$$

DOUBLE ANGLE:

$$\begin{aligned}\sin(2\theta) &= 2 \sin \theta \cos \theta \\ \cos(2\theta) &= \cos^2 \theta - \sin^2 \theta = 2\cos^2 \theta - 1 = 1 - 2\sin^2 \theta\end{aligned}$$

SUM/DIFFERENCE (provided on formula sheet but useful to recognize):

$$\begin{aligned}\sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \\ \cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B\end{aligned}$$

EVEN/ODD: $\cos$ is even ($\cos(-\theta) = \cos \theta$); $\sin$ & $\tan$ are odd

Inverse Trig Functions — restricted ranges [5]

$$\begin{aligned}\arcsin(x) &= \sin^{-1}(x): \text{ Range } [-\pi/2, \pi/2] && \text{(Q4, Q1)} \\ \arccos(x) &= \cos^{-1}(x): \text{ Range } [0, \pi] && \text{(Q1, Q2)} \\ \arctan(x) &= \tan^{-1}(x): \text{ Range } (-\pi/2, \pi/2) && \text{(Q4, Q1)}\end{aligned}$$

CRITICAL: $\arcsin(\sin \theta)$ only equals θ when θ is IN the range above.
 Always verify the quadrant.

Solving Trig Equations [5]

- Isolate the trig function first. $2 \sin x - 1 = 0 \Rightarrow \sin x = 1/2$.
- Find ALL solutions in the given interval (often $[0, 2\pi)$). For $\sin x = 1/2$: $x = \pi/6$ AND $x = 5\pi/6$ ($\sin$ is + in Q1 AND Q2).
- General solution (no interval given): add $+ 2\pi k$ for $\sin/\cos$ and $+ \pi k$ for $\tan$.
- Quadratic-in-trig: $2 \sin^2 x - \sin x - 1 = 0 \rightarrow$ factor as $(2 \sin x + 1)(\sin x - 1) = 0 \rightarrow$ solve each piece.

UNIT 3 | SINUSOIDAL MODELING + POLAR FUNCTIONS

Sinusoidal Functions — the 4 parameters [5]

General form: $f(x) = a \cdot \sin(b(x - h)) + k$
 $f(x) = a \cdot \cos(b(x - h)) + k$

a = AMPLITUDE = $(\max - \min) / 2$
 k = MIDLINE / vertical = $(\max + \min) / 2$
 b = ANGULAR FREQUENCY $\rightarrow$ PERIOD = $2\pi / b$
 h = PHASE SHIFT (horizontal shift, right = $+h$)

Conversions: PERIOD given $\rightarrow b = 2\pi / \text{period}$
 FREQUENCY given (cycles/unit) $\rightarrow b = 2\pi \cdot \text{frequency}$

Building a Sinusoidal Model from Data — the 5-step recipe [5]

This is the score-defining move on FRQ 3 (Periodic Modeling). Memorize this 5-step recipe.

Step 1: Find **midline** $k = (\max + \min) / 2$.

Step 2: Find **amplitude** $a = (\max - \min) / 2$.

Step 3: Find **period** (one full cycle), then $b = 2\pi / \text{period}$.

Step 4: Choose $\sin$ or $\cos$. Use $\cos$ if the function starts at **max** at $x = 0$ (or $x = h$). Use $\sin$ if it starts at the midline rising.

Step 5: Find the **phase shift** h — the x -value where the chosen reference behavior occurs (where $\cos = \max$, or $\sin = \text{midline rising}$).

[5] Always check by evaluating your model at a known point (e.g., the given max or min) and verifying it matches.

Worked Example — building a sinusoidal model [5]

Problem: A Ferris wheel reaches max height of 50 ft, min of 10 ft, completes a cycle every 4 minutes, and is at maximum at $t = 1$.

Step 1: Midline $k = (50 + 10)/2 = 30$.

Step 2: Amplitude $a = (50 - 10)/2 = 20$.

Step 3: Period = 4, so $b = 2\pi/4 = \pi/2$.

Step 4: Use $\cos$ since at max at $t = 1$.

Step 5: Cosine reaches max at its argument = 0, so $(\pi/2)(t - h) = 0$ at $t = 1 \rightarrow h = 1$.

Final model: $H(t) = 20 \cos((\pi/2)(t - 1)) + 30$

Polar Coordinates — `(r, θ)` system [5]

Polar $\rightarrow$ Rectangular: $x = r \cos \theta$ $y = r \sin \theta$
 Rect $\rightarrow$ Polar: $r^2 = x^2 + y^2$ $\tan \theta = y / x$

Note: r can be negative (point in opposite direction);
 θ is typically given in radians, in $[0, 2\pi)$ for principal value.

Common Polar Graphs — recognize them by form [5]

Polar equation form	Type of curve	Key feature
$r = a$	Circle	Radius a , centered at origin
$r = a \cos \theta$ or $r = a \sin \theta$	Circle	Diameter a , passes through origin
$r = a + b \cos \theta$ ($a, b > 0$)	Limaçon	Cardioid if $a = b$ (heart shape); inner loop if $a < b$; dimpled if $a > b$ but $a < 2b$
$r = a \sin(n\theta)$ or $r = a \cos(n\theta)$	Rose curve	n petals if n is odd; $2n$ petals if n is even
$r = a \cdot \theta$	Spiral (Archimedean)	Radius grows linearly with angle
$r^2 = a^2 \cos(2\theta)$	Lemniscate	Figure-8 shape

Average Rate of Change of `r` over `[θ<sub>1</sub>, θ<sub>2</sub>]` [5]

Avg rate of change of r over $[\theta_1, \theta_2] = (r(\theta_2) - r(\theta_1)) / (\theta_2 - \theta_1)$

Interpretation: tells you how fast the radius is changing PER UNIT ANGLE.
 Positive $\Rightarrow$ moving outward; negative $\Rightarrow$ moving inward.

PAGE 7 | FRQ STRATEGY · templates for the 4 fixed FRQ types

UNIVERSAL FRQ DISCIPLINE — non-negotiable for a 5 [5]

- **Show the formula setup** before plugging in numbers. "Average rate of change = $(f(b) - f(a))/(b - a) = (12 - 4)/3 = 8/3$."
- **Use proper notation.** $f^{-1}(x)$, NOT "the inverse" in words. $\lim$ is NOT used in precalc — but $f(x) \rightarrow \infty$ is acceptable end-behavior notation.
- **Carry units through context problems.** "In the context of the problem, this represents 8/3 dollars per year."
- **Answer in complete sentences when prompted.** "Justify" / "Explain" / "Interpret in the context of" $\rightarrow$ write a sentence, not just a number.
- **Don't erase aggressively.** Cross out cleanly with a single line — graders will award the right answer if visible.

FRQ 1 — FUNCTION CONCEPTS (calculator)

Format: Given a function f (graph, table, or equation). 4 parts asking for various characteristics.

Common asks (and templates):

- "Find all values of x where $f(x) = 0$." $\rightarrow$ Use **calc zero feature** + report. " $f(x) = 0$ at $x = 1.273$ and $x = 4.812$."
- "Find the average rate of change over $[a, b]$." $\rightarrow$ Setup + plug-in: " $(f(b) - f(a))/(b - a) = \dots$."
- "On what interval is f increasing?" $\rightarrow$ State interval **with reasoning**: " f is increasing on $(2, 5)$ because the graph rises throughout this interval."
- "Describe end behavior." $\rightarrow$ "As $x \rightarrow \infty$, $f(x) \rightarrow \infty$. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ because the polynomial has odd degree with positive lead coefficient."

FRQ 2 — MODELING NON-PERIODIC (calculator)

Format: Real-world data + scenario. Choose model family, fit, interpret.

Score-earning template:

- **Part A (recognize family):** "Because the data exhibits **constant ratios** between successive y -values when x increases by 1, the data is best modeled by an **exponential function**."
- **Part B (build model):** Use calc regression OR algebraically. Write final equation: " $P(t) = 250 \cdot (1.08)^t$."
- **Part C (interpret parameters):** " $a = 250$ represents the initial population in year 0, and $b = 1.08$ indicates an **8% annual growth rate**."
- **Part D (predict / extrapolate):** Use model. " $P(20) = 250 \cdot (1.08)^{20} \approx 1165$. According to the model, the population in year 20 will be approximately 1,165."

FRQ 3 — MODELING PERIODIC (no calculator)

Format: Sinusoidal scenario (Ferris wheel, tide, daylight, temperature). Build the model and use it.

Score-earning template — apply the 5-step recipe from Page 6:

- **Part A (find amplitude + midline):** "Amplitude = $(\max - \min)/2 = \dots$ Midline = $(\max + \min)/2 = \dots$ "
- **Part B (find period and b):** "Period = ... so $b = 2\pi / \text{period} = \dots$ "
- **Part C (write the model):** Choose sin or cos based on starting position. " $f(t) = a \cdot \cos(b(t - h)) + k$."
- **Part D (apply / interpret):** Plug in a specific time. **Always interpret in context.**

FRQ 4 — SYMBOLIC MANIPULATIONS (no calculator)

Format: Pure algebra — solve, simplify, prove identities, manipulate.

Score-earning template:

- **Show every step.** Even "obvious" simplifications. The rubric scores **process** as well as result.
- **Use standard algebraic notation.** Equal signs aligned vertically when stacking equations.
- **For identity proofs:** Start from one side, manipulate to match the other. **Never work both sides simultaneously** — that's circular.
- **Check work by substitution** if time allows. For solved equations, plug in answer + verify.

PAGE 8 | TRAPS, 4-vs-5 LADDER & TEST-DAY DISCIPLINE

THE TOP-12 TRAPS — where strong students slip [5]

- **Calculator in DEGREE mode** when the problem is in radians (default for AP Precalc). **Check at the start of every section.**
- **Wrong direction on horizontal shift.** $f(x - 3)$ shifts RIGHT 3, not left. Sign is opposite of what "feels right."
- **Confusing inverse with reciprocal.** $f^{-1}(x)$ is NOT $1/f(x)$. $\sin^{-1}(x)$ is NOT $\csc(x)$.
- **Forgetting to check extraneous solutions** in log equations. Logs require positive arguments.
- **Wrong end behavior on rational functions.** Compare degrees of numerator and denominator before answering.
- **Missing solutions on trig equations.** $\sin x = 1/2$ has TWO solutions in $[0, 2\pi)$: $\pi/6$ AND $5\pi/6$.
- **Mixing up amplitude and period.** Amplitude affects vertical stretch (a). Period affects horizontal compression (b).
- **Wrong choice of sin vs cos** in sinusoidal modeling. Cos starts at max; sin starts at midline rising.
- **Forgetting $+2\pi k$** when asked for general solutions to trig equations.
- **Inverse trig range errors.** $\arcsin$ outputs $[-\pi/2, \pi/2]$; $\arccos$ outputs $[0, \pi]$. Check the quadrant.
- **Skipping algebra steps on FRQ 4.** Even if you can do it in your head, **write it down.**
- **Reporting answers without context** on FRQ 2 (modeling). The rubric requires interpretation.

THE 4-vs-5 LADDER — what the gap looks like at the answer level

A 4-scorer writes...	A 5-scorer writes...
" $x = 4$ "	"Setting $f(x) = 0$: $x^2 - 5x + 4 = 0 \rightarrow$ factoring: $(x - 4)(x - 1) = 0 \rightarrow x = 1$ or $x = 4$. The function has zeros at $x = 1$ and $x = 4$, where the population returns to zero. "
"The function is exponential."	"Because the data shows constant ratios (each y-value is approximately 1.08 times the previous), the appropriate model is exponential : $f(x) = a \cdot b^x$. Using the first two points to solve for a and b..."
Identifies asymptotes correctly	Identifies asymptotes AND explains the algebraic reason (e.g., "vertical asymptote at $x = 3$ because the denominator equals zero there but the numerator does not")
Builds sinusoidal model with correct numbers	Builds the model AND interprets each parameter in context ("the amplitude of 20 represents the maximum deviation of the temperature from the average")
States interval of increase	States interval of increase AND justifies with reference to graph features or rate of change
Solves trig equation, gives one answer	Gives all solutions in the requested interval, AND adds $+ 2\pi k$ if asking for general solution
Reports $f(20) = 1165$	Reports $f(20) = 1165$ AND interprets : "according to the model, the population in year 20 will be approximately 1,165 — though this is an extrapolation and assumes the trend continues"

PHRASES THAT EARN POINTS [5]

For interpretation: "In the context of the problem, this represents..." / "This means that for every unit increase in x, the function increases by..."

For justification: "because the rate of change is positive throughout this interval" / "because the leading coefficient is positive and the degree is odd"

For modeling: "The data exhibits [constant differences / constant ratios / constant 2nd differences], so the appropriate model is [linear / exponential / quadratic]."

For trig reasoning: "In the interval $[0, 2\pi)$, the equation $\sin x = c$ has solutions in **Q1 and Q2** because sine is positive in those quadrants."

For end behavior: "As $x \rightarrow \infty$, $f(x) \rightarrow$ [value or $\pm\infty$] because [degree comparison / asymptotic reason]."

TEST-DAY EXECUTION CHECKLIST

Before Section 1: ✓ Calculator in **RADIAN** mode ✓ Calculator batteries fresh ✓ Spare calc if allowed ✓ Pencil + extra

For a 4 — bag ALL of these: ✓ MCQ Part A: 19+ correct (be efficient on no-calc; eliminate aggressively) ✓ MCQ Part B: 8+ correct (use calc liberally for arithmetic) ✓ Each FRQ: complete every part — even partial work earns points ✓ NEVER leave anything blank

For a 5 — add these on top: ✓ MCQ Part A: 21+ correct ✓ MCQ Part B: 9+ correct ✓ Every FRQ: **show formula setup** before computing ✓ Use **proper notation** throughout (f^{-1} , $\rightarrow$, intervals) ✓ **Interpret in context** on every modeling problem ✓ Justify with reasoning when asked ✓ Check answers by substitution if time allows

THE ONE THING THAT MATTERS MOST [5]: SHOW WORK + INTERPRET. AP Precalc grades process and interpretation as much as final answers. After every numerical answer, ask: "Did I show how I got there, and did I explain what it means in context?" If yes to both — you're at a 5.

If you do nothing else: memorize the unit circle (Page 5), master the 5-step sinusoidal recipe (Page 6), and use the FRQ templates (Page 7). That's how you get the score you're targeting.

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